

Kuroshio Current regional model validation

Pending alignment draft

Field	Entry
Region	Kuroshio Current, LME_049. The source spans coastal and offshore western North Pacific blocks. Regional workbook
Catch source	Sea Around Us v50.1, 1950–2019 reconstructed catch. Reference: 2019 landings; all 253 reporting labels, including zero catch. Original catch
Selected article	Watari et al. (2019). Ecosystem modeling in the western North Pacific using Ecopath, with a focus on small pelagic fishes. DOI 10.3354/meps12508. Main PDF verified; supplementary text only partially readable. Original article Partial supplement
Other known articles	Chen et al. (2025), Study in the ecosystem structure and trophodynamics in the Kuroshio-Oyashio Extension area. One 25-group 2023 offshore survey model; conditional GE, TE and Egestion diagnostics FAIL. Not selected. Main article
Selected model	49_2019201301_Watari_Detritus_Pooled_Experiment_(2013). One static 2013 assumption scenario: 35 consumers, 3 spatial phytoplankton and pooled detritus; 39 source groups plus computational import 40. Source area 913,102 km ² . Accepted model
Other models from the same article	Original 41-group Watari 2013 model retains three spatial detritus pools. Missing between-pool routing blocks its source-faithful SPPR load. OYC, KC and OF form one linked model; they are not three independent baselines. Original model
Selection rationale	Article — User preferred the finer 41-group source structure. Model — User selected detritus-only pooling because source routing is unavailable. Cross-block mixing and full internal retention remain assumptions; selection does not establish scientific validity.
Model extraction	Accepted pooling sums total-area detritus B to 44.12 without second habitat weighting. Canonical non-detritus inputs remain exact. Retained loader changes include 83 normalized diet cells, missing/default GS/catches, seabird B=1 and PB=QB=0, and solved BA. Pooled detritus BA +1,021.626599 is computational closure, not measured change. Source and runtime differences
SPPR calculation	[MANUAL — Researcher: calculation choices and reasons] [Excluded groups, scope and relevant values; other adjustments]

Field	Entry
GE diagnostics	No negative SPPR entries across the basal-source columns. rho_living=0.33343023; b=0.04017288; pooled Detritus 39 SPPR=1.40229805. Full source matrices
TE diagnostics	Detritus (pooled OYC KC OF; experiment) 39 → Seabirds (all) 3; Phytoplankton (OYC) 36 → Seabirds (all) 3; Phytoplankton (KC) 37 → Seabirds (all) 3; Phytoplankton (OF) 38 → Seabirds (all) 3. TE FAIL remains. rho_living=0.54627838; b=0.00000000; pooled Detritus 39 SPPR=0.64654301. b=0 is the TE convention. Full source matrices
Geographic fit	A. Region covered by study: approximately 30–36%. B. Study covered by region: approximately 49–51%.
Temporal fit	Static 2013 coefficients and fixed source-catch/biomass proportions transfer across 1950–2019 whole-LME landings, catch and discards. Reference 2019 is six years later. This does not reconstruct annual ecosystem dynamics or observed spatial catch composition.
Other	GE and With Egestion retain threshold OK with strict SPPR balance false. All 600 source-matrix entries were checked, including unfished groups. Near-singular TE and existing MC 0/100 accepted remain; the cause is deferred. Partial later supplement and habitat proxies limit membership. Global diagnostic returned no object. Diagnostics and limits
Open issues and next action	[Researcher: open issues and next actions]
Review and reproducibility	Researcher name: [name] review date: [date]

Taxon mapping and coverage

Reference: 2019 landings; all source scope, all exact reporting labels, no group filter.

39 source groups (+1 computational import); 253 catch taxa.

Simple-chain PPR totals 180,683,377.69 t C in 2019.

24 labels lack a classic coefficient/TL; all have zero 2019 landings and zero annual contribution. Unknown coefficients and TL remain ?.

Method: protected independent simple trophic-chain coefficients.

[All 253 taxon mappings](#) and [descriptive Sources sheet](#)

Overall confidence	Taxa	Catch percentage	PPR percentage
High	15	35.2337%	29.2963%
Medium	143	31.8095%	51.7575%
Low	1	0.0000%	0.0000%
Very low	94	32.9568%	18.9463%
Unresolved	0	0.0000%	0.0000%

The summary uses the weaker required membership or allocation confidence. The following tables show each component separately.

Group assignment rules

Plain-language rule	Confidence	PPR percentage
Exact species explicitly named by the selected source.	High	27.3207%
Assumed eligible source family or spatial/depth set.	Medium	23.5701%
Supported ecological extension to a represented guild.	Medium	21.6600%
Closest represented habitat/taxonomic analogue, with mismatch.	Very low	12.6790%
Supported extension from named source representatives.	Medium	6.5274%
Broad reporting union approximated across compatible pools.	Very low	6.2673%
Authoritatively verified source-to-reporting synonym.	High	1.9756%
Partial ecological fit with a material source mismatch.	Low	0.0000%

Membership evidence: [main Table 1 and Section 2.3](#), [partial Supplement 1](#), [exact taxonomy and traits](#) and [all decisions](#)

Allocation weight rules

Plain-language rule	Confidence	PPR percentage
One reviewed group; no numerical split.	High	54.5095%
Complete whole-domain biomass after incomplete/censored catch.	Medium	29.6102%
Complete printed source Landings proportions, used as a proxy.	Medium	15.8803%

Weights and assumptions: complete 2013 source Landings first; otherwise complete total-area biomass. Censored <0.01 and dash/missing catch never become measured zero. No second habitat fraction is applied. Proxies assume composition/catchability and remain fixed across years and catch bases. No usable source catch time-series or observed caught-stage split was recovered. Each table partitions the same simple-chain PPR. [Complete candidate vectors](#)

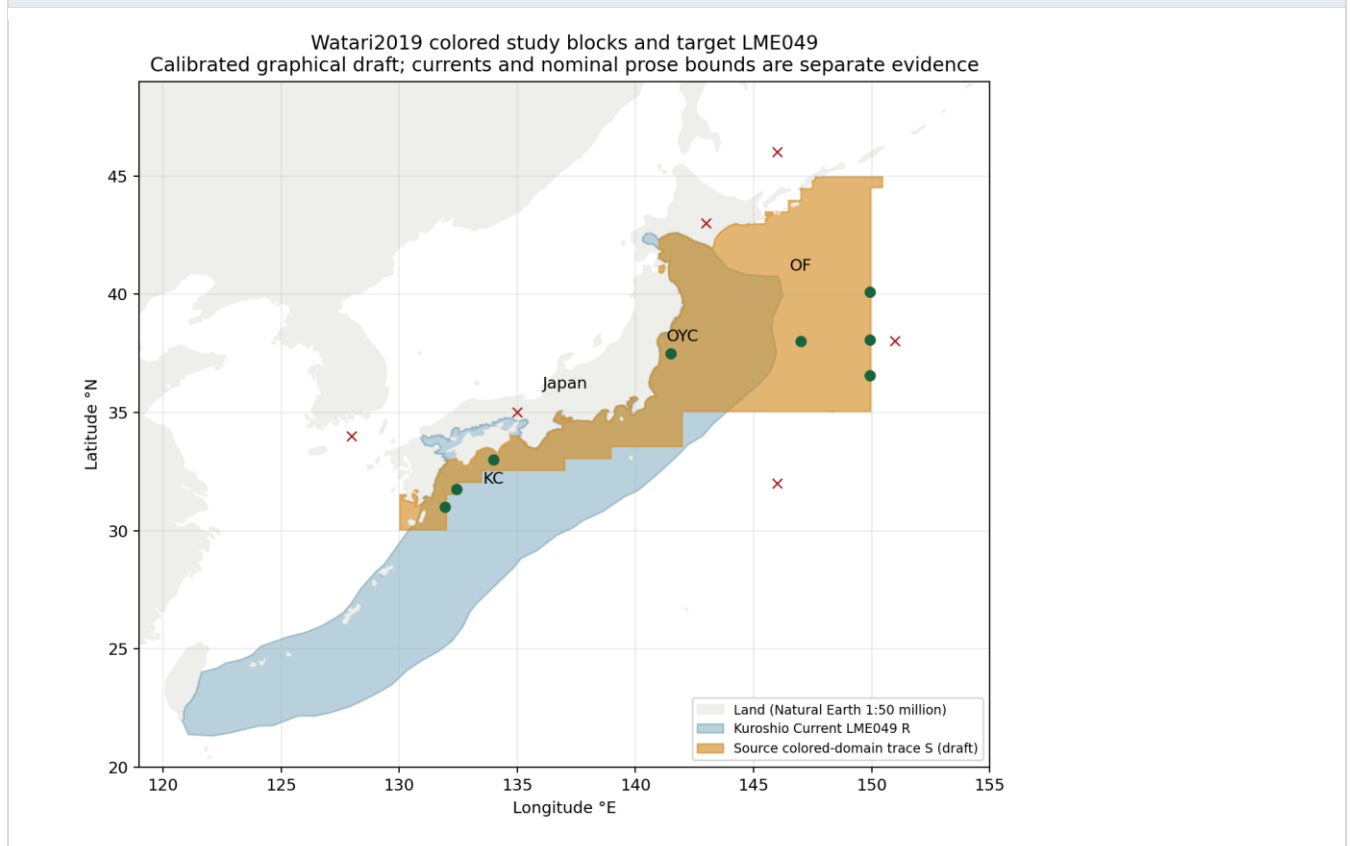
Unresolved: none. Very low: 94 labels account for 32.9568% of landings and 18.9463% of simple-chain PPR. They include broad bony/invertebrate pools and unrepresented reef/deep fish, jellyfish and benthic cephalopods. Broad Sepiida includes bobtail depth habitats; narrower octopus/cuttlefish labels use benthos analogues. Very low all-source model PPR shares: GE 13.71%; TE 24.49%; With Egestion 15.10%; TE remains failed. [Exact flagged appendix rows](#) | [Largest method contributors](#)

Geographic evidence screenshots

R is the Sea Around Us Kuroshio Current LME. S is the calibrated colored KC/OYC/OF footprint in original Figure 1. This graphical reconstruction has Low confidence; enclosing 30–50°N/150°E prose does not establish a full rectangular study polygon.

Region area

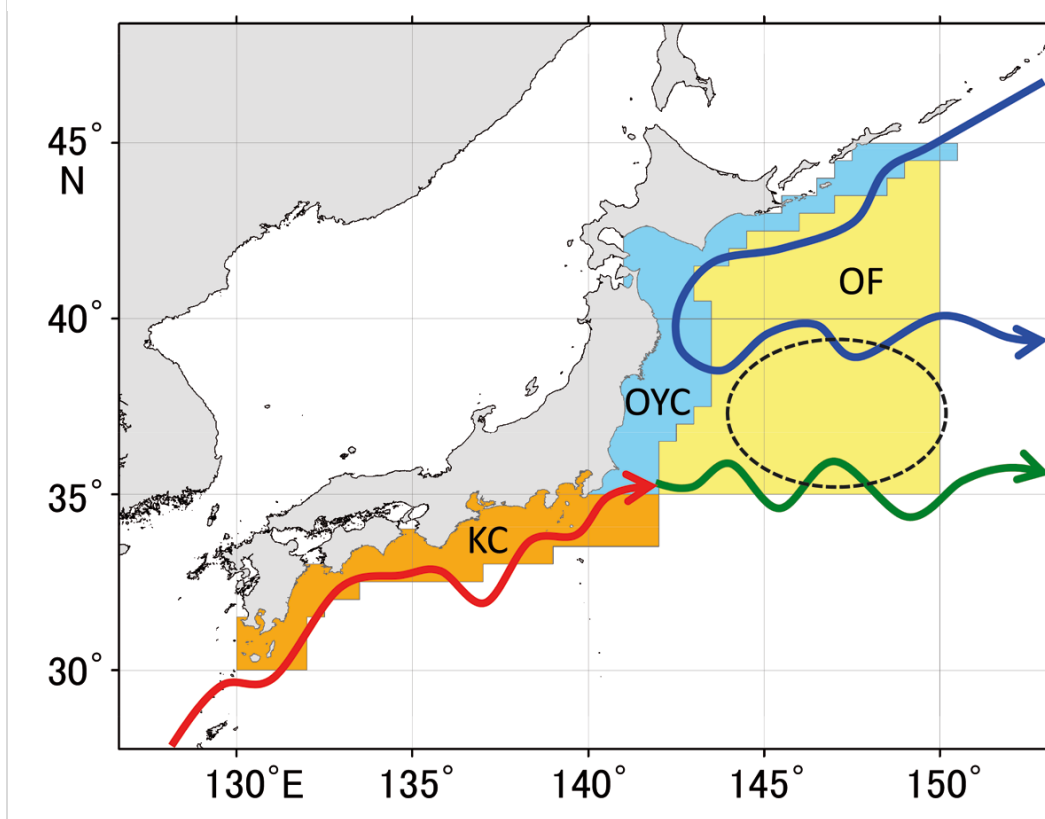
Target LME049 and calibrated source-colored domain



Target R: 1,320,443 km². Central trace S: 887,821 km²; intersection 448,167 km². Approximate A/B ranges include $\pm 0.10^\circ$ graphical sensitivity. Fourteen ordered water/land controls pass. Generalized coastline and schematic grid boundaries limit accuracy. [Target polygon](#) | [Calibrated trace](#)

Study area in the article

Original Figure 1 with coastal and offshore study blocks



Original main PDF 3/printed 296. KC and OYC are coastal water pools; offshore OF is the colored stepped wedge. Current arrows and the transition-zone ellipse are overlays, not water exclusions. Published source area 913,102 km² is separate from traced area. No geographic multiplier is applied. [Original Figure 1](#) | [Assessment and controls](#)

Figure 1 and methods give KC 186,220 and OYC 186,128 km²; Table 4 reverses those labels while preserving the total. The discrepancy is retained as source context without changing accepted model inputs.